

Thien-An Tran

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EDUCATION

University of Washington | BSc. Computer Science

2022 – 2026

Relevant Coursework: Cryptography, Secure Coding Principles, Data Structures & Algorithms, Machine Learning, AI & Knowledge, Machine Organization, Database Systems Design, Game Simulation and Design

TECHNICAL SKILLS

Languages | Java, Python, JavaScript, C#, C++, SQL, C, Erlang, Assembly

Technologies | Cryptography (SHA-3, SHAKE, NIST FIPS 202, ECIES, Schnorr), PyTorch, OpenCV, NumPy, React, Supabase, PostgreSQL, Cloudflare, Microsoft Azure, Node.js, Git, Firebase

PROFESSIONAL EXPERIENCE

Full Stack Developer | Vietnamese Martyr's Parish

2024 – 2026

- Developed a secure document management system using React and Microsoft Azure Blob Storage
- Implemented access controls and data integrity measures for 500+ sensitive user records
- Designed and maintain a SQL database of 500+ user records with optimized queries and strict data integrity enforcement for sensitive personal documentation
- Implemented security best practices including parameterized queries, access controls, and privacy-compliant data handling for records management

Web Developer | PROstrada

2024 – 2025

- Designed and developed interactive web applications in React by translating client requirements into functional software, ensuring performance and responsiveness standards were met
- Partnered with non-technical stakeholders to translate operational requirements into software designs, aligning technical solutions with business and user needs
- Maintained and enhanced software products iteratively based on stakeholder feedback
- Managed requirements changes throughout the development lifecycle

PROJECTS

Plately | *React, Supabase, Claude API, Cloudflare*

2026

- Built a full-stack AI-powered nutrition-tracking PWA enabling food logging via photo, barcode, voice, and natural language, with offline-first sync across devices
- Designed a secure serverless API architecture using Cloudflare Pages Functions to inject secret keys server-side, ensuring no third-party API keys are ever exposed to the client
- Integrated Anthropic Claude vision and NLP models with structured-output parsing and multi-provider failover across five nutrition APIs for graceful degradation
- Implemented a Postgres database with Row-Level Security via Supabase and a service-worker caching strategy for full offline functionality

Cryptography Library | *Java*

2026

- Implemented SHA-3-256, SHA-3-512, SHAKE-128, and SHAKE-256 from scratch in Java conforming to NIST FIPS 202, verified against official test vectors
- Designed symmetric encryption and decryption system using SHAKE-128 as a stream cipher with a 128-bit random nonce and passphrase-derived key, and a SHAKE-based message authentication code for tamper detection
- Implemented NUMS ed-256-mers* Edwards elliptic curve arithmetic in Java including point addition, scalar multiplication, and subgroup validation
- Built ECIES public-key encryption scheme using SHAKE-based KDF, stream cipher, and MAC tag verification for authenticated encryption
- Implemented Schnorr digital signatures with deterministic nonce generation for message signing and verification

Food Image Classifier | *Python, PyTorch*

2026

- Trained a convolutional neural network on the Food-101 benchmark dataset (101 categories, 101,000 images) achieving 80% top-1 accuracy on the held-out test set
- Applied transfer learning from a pre-trained model with data augmentation and fine-tuning to improve generalization across diverse food categories
- Built the classifier to power automated calorie estimation in a nutrition-tracking application

AI Autonomous Agent | *Python*

2025

- Designed and implemented a multi-agent autonomous navigation system in Python integrating adversarial search with probabilistic Bayesian reasoning under sensor uncertainty
- Implemented and documented five pathfinding algorithms (BFS, DFS, UCS, Greedy, A*) analyzing optimality and efficiency tradeoffs across constrained grid environments
- Built a Bayesian probabilistic reasoning engine handling false-positive and false-negative sensor readings, maintaining belief distributions updated via posterior inference